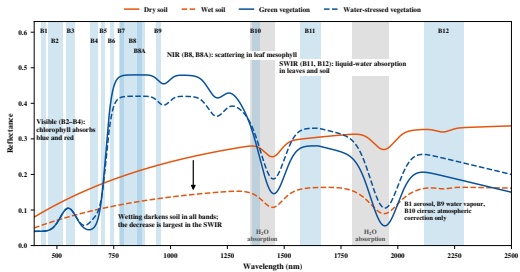
Motivation Different ways of obtaining soil moisture

Source	How SM is obtained	Depth	Main limitation
Microwave (SMAP, SMOS, ASCAT, S1)	invert: σ^0 or $T_B = f(\text{SM}, \text{roughness}, \text{vegetation})$	top $\sim$5 cm; informs part of the root zone (Feldman et al., 2023)	shallow; assumes vegetation and roughness; coarse (9–40 km) except S1
Optical / thermal (S2, Landsat, MODIS)	empirical relations with surface properties: reflectance and indices; LST–vegetation space (e.g. triangle method, TVDI)	skin / surface	indirect, site-specific calibration; clouds; canopy hides the soil
Land-surface models	$d\theta/dt = P - ET - Q - D$, with parameters p	full profile	forcing errors, soil and vegetation parameters, process representation (Koster et al., 2009; Seneviratne et al., 2010)
Reanalysis (ERA5-Land)	LSM driven by reanalysis meteorology	full profile	model assumptions dominate deep layers with few observational constraints (Li et al., 2021; Miguez-Macho and Fan, 2021)
In situ (ISMN)	direct probe measurement	several depths	point scale; sparse, uneven global coverage
Data-driven (ML / DL)	$\text{SM} = f_\theta(\text{inputs})$, learned from in-situ labels; no ad-hoc process assumptions	any depth with labels	needs enough, representative sites; generalisation?; hard to interpret?

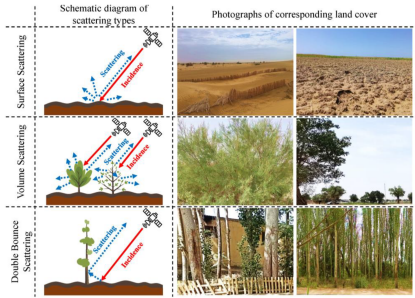
Motivation Multimodal remote sensing

1. Optical: reflectance spectrum



- Dry and wet soil reflect differently: wetting darkens soil in every band, most in the SWIR (B11, B12)
- Green and water-stressed vegetation reflect differently: stress lowers NIR and raises SWIR

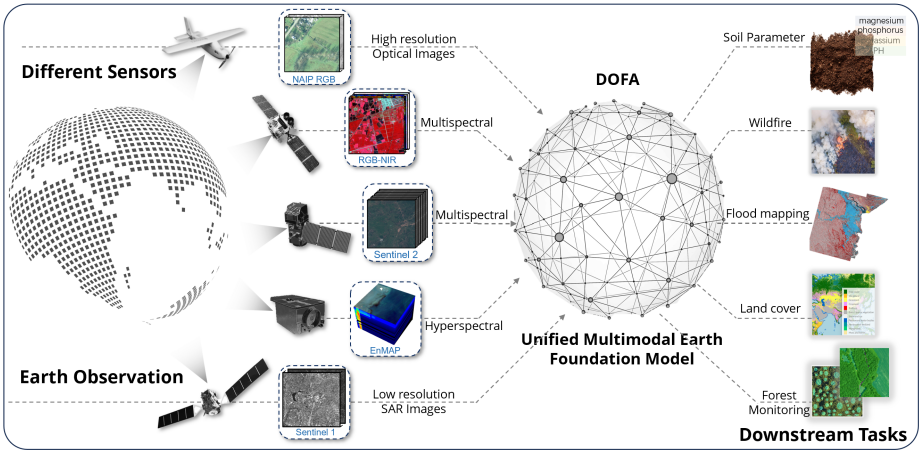
2. SAR: scattering mechanisms



Nurmemet et al. (2025), *Sensors* 25(8), 2512, doi:10.3390/s25082512

- Bare soil: surface scattering, set by moisture (dielectric) and roughness
- Canopy: volume and double-bounce scattering mask the soil signal

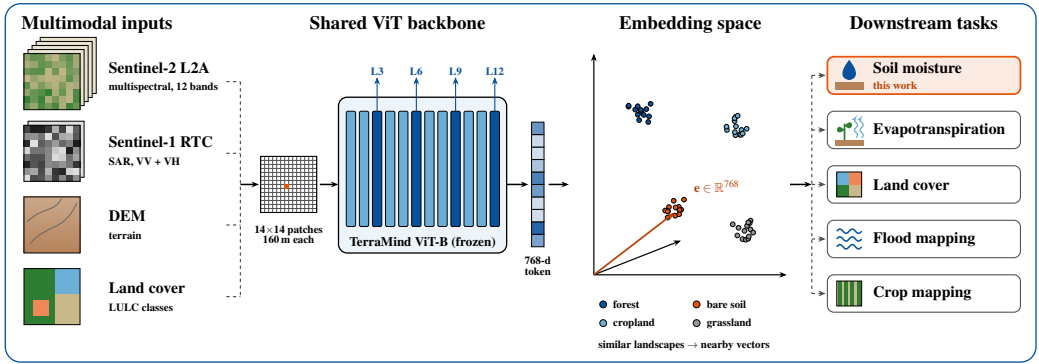
Motivation Geospatial foundation models (GFMs)



DOFA: Xiong et al. (2024), Neural plasticity-inspired multimodal foundation model for Earth observation, arXiv:2403.15356

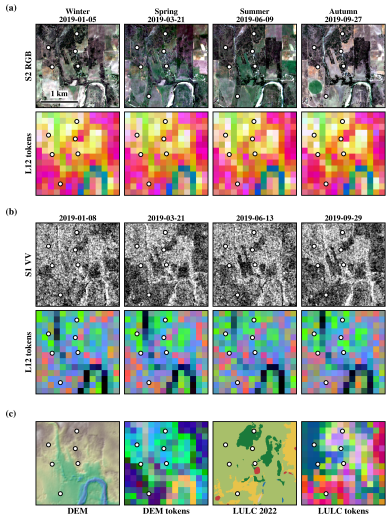
Once pre-trained on multimodal, unlabelled data, its representation works for many downstream tasks

Motivation TerraMind embeddings for soil moisture



Frozen TerraMind ViT-B turns each 160 m patch of S2, S1, DEM and LULC into a 768-d embedding; the same embedding can serve many tasks; in this case, **soil moisture**.

Motivation Example: TerraMind embeddings at one site (TxSON, 2019)



- (a) S2 RGB in four seasons and its L12 tokens; (b) S1 VV and its tokens; (c) DEM, LULC and their tokens
- Each 160 m patch $\rightarrow$ one 768-d vector; colours show its first 3 principal components (one basis per row, each panel centred on its own mean)
- Tokens follow fields, woodland and the river: the landscape structure is preserved at 160 m
- The within-tile token *pattern* is almost the same in all four seasons: it describes *what the land is* more than *how wet it is today*